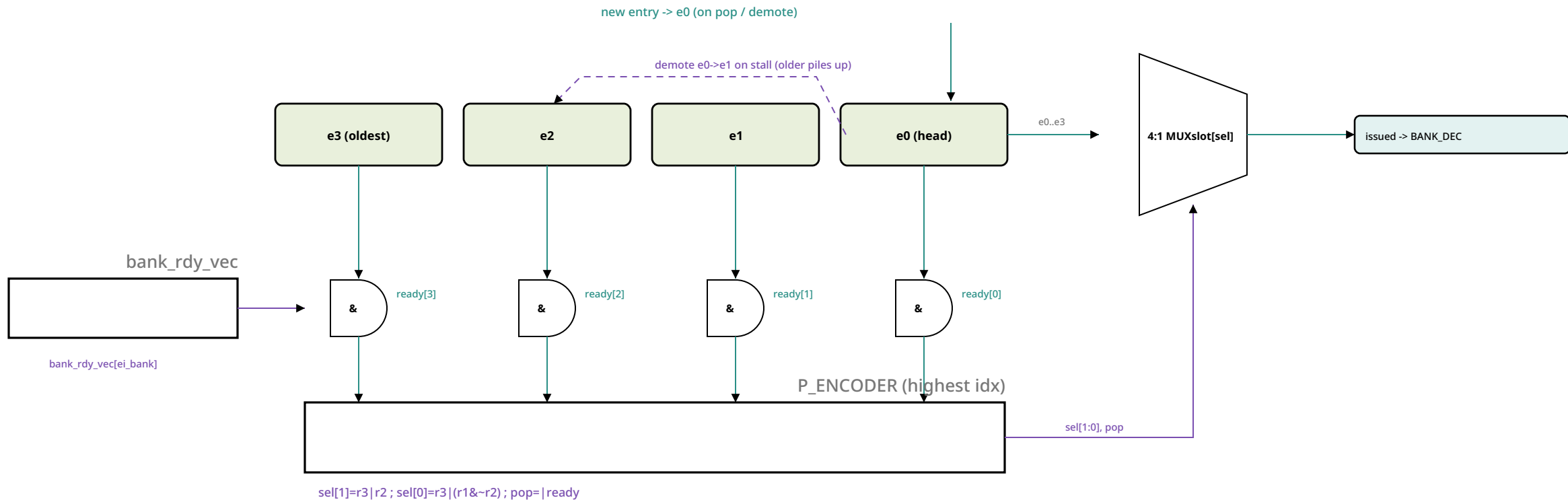


LOOKAHEAD (final) - fixed 4:1 mux + binary highest-idx p-enc + demote



$\text{ready}[i] = ei_vld \& \text{bank_rdy_vec}[ei_bank]$. Binary p-enc picks HIGHEST set idx (oldest; demote pushes stalled heads up) -> $\text{sel}[1:0]$; $\text{pop} = | \text{ready}$. One fixed 4:1 mux pops $\text{slot}[\text{sel}]$ -> freed stalled ei beats fresh $e0$. On pop: compact (newer shift up, $e0 \leftarrow \text{new}$).